Atomics and the Memory Model

Atomic operations, compare-and-swap, memory orders and happens-before, the ordering patterns that matter, and thread-safe data structures from fine-grained locks to lock-free. Builds on Multithreading.

Atomic operations and `compare_exchange`

`std::atomic<T>` makes operations on **one variable** indivisible, usually without blocking. The increment is now one step.

```
#include <atomic>

std::atomic<int> counter{0};
// one indivisible read-modify-write
counter++;
counter.fetch_add(1);                    // same thing, explicit
int v = counter.load();                  // atomic read
counter.store(5);                        // atomic write
// write 9, get the previous value, all in one step
int old = counter.exchange(9);

// a very common use: a shutdown flag
std::atomic<bool> stop{false};
```

USE AN ATOMIC FOR	DO **NOT** USE IT FOR
a counter, flag, swapped pointer, statistics	an invariant over **several** variables: use a mutex

Compare-and-swap (CAS)

`compare_exchange(expected, desired)`: "if the value still equals `expected`, set it to `desired`; otherwise load the current value into `expected` and fail". It is the building block of every lock-free update.

```
int expected = x.load();
while (!x.compare_exchange_weak(expected, expected * 2)) {
        // someone changed x: `expected` now holds the current value, so
    // recompute and retry
}
```

	`COMPARE_EXCHANGE_WEAK`	`COMPARE_EXCHANGE_STRONG`
spurious failure	**may** fail even when the value matched	fails only if the value really differed
use	inside a retry **loop** (cheaper on some CPUs)	a single attempt, no loop

Compiled: a CAS with a stale `expected` returned false and refreshed `expected` to the current value (99).

`atomic_flag` and a spinlock

```
struct Spinlock {
    // C++20: a plain `std::atomic_flag f;` starts clear, and
    // ATOMIC_FLAG_INIT is deprecated
    std::atomic_flag f = ATOMIC_FLAG_INIT;
    void lock()   { while (f.test_and_set(std::memory_order_acquire)) { /*
spin */ } }
    void unlock() { f.clear(std::memory_order_release); }
};
```

`atomic_flag` is the one type guaranteed lock-free. Four threads each doing 50,000 increments under this spinlock reached exactly 200,000 (compiled). A spinlock burns a core while waiting: fine for tiny critical sections on dedicated cores, bad when the lock holder can be descheduled (use a mutex).

> **VOLATILE IS NOT THREAD-SAFE** It only stops the compiler optimising an access away. No atomicity, no ordering. Use `std::atomic`.

- `is_lock_free()` says whether the type is done with real atomic instructions; big structs fall back to a hidden lock.
- **ABA problem:** a CAS can succeed when the value went A to B and back to A, although the world changed meanwhile. Fix with a version counter next to the value, or hazard pointers.

Memory orders and happens-before

Compilers and CPUs **reorder** memory operations, and each core has buffers and caches, so one thread's write may become visible to another thread *later*, or *after* a write that came after it. A plain `int` gives no guarantee. Atomics let you say how much ordering you need.

The one rule to learn: release and acquire

```
int data = 0;
std::atomic<bool> ready{false};

// producer thread
data = 42;
ready.store(true, std::memory_order_release);

// consumer thread
while (!ready.load(std::memory_order_acquire)) { }
std::cout << data;                        // guaranteed to print 42
```

- A **release store** publishes: every write the thread made *before* it becomes visible to any thread that reads the stored value with an **acquire load**.
- The release/acquire pair **synchronizes-with**, and that creates a **happens-before** edge: `data = 42` happens before the consumer's read, so the read is not a data race and sees 42.
- **happens-before** is the formal "must be visible" relation. Within one thread it is program order; across threads it comes only from synchronization (atomics, mutexes, `join`, futures).

What the consumer can print in the publish-with-a-flag example above, for each pair of memory orders:

`ready.store(true, ...)`	`ready.load(...)`	WHAT THE CONSUMER PRINTS
`release` (or `seq_cst`)	`acquire` (or `seq_cst`)	always **42**
`relaxed`	`acquire`	may print **0**: the flag can become visible before the data
`release`	`relaxed`	may print **0**: nothing guarantees the data is visible after the load
`relaxed`	`relaxed`	may print **0**

The orders

ORDER	GUARANTEE	USE
`relaxed`	only atomicity (and a single total order for *that* variable)	counters and statistics nobody synchronizes on
`acquire` (loads)	nothing after it moves before it; sees what the matching release published	reading a flag or pointer that publishes data
`release` (stores)	nothing before it moves after it; publishes earlier writes	setting that flag or pointer
`acq_rel` (read-modify-write)	both at once	`fetch_sub` on a reference count, a lock hand-over
`seq_cst` (default)	acquire/release **plus** one global order of all `seq_cst` operations that every thread agrees on	when in doubt: it is the only order that is easy to reason about

> `X86 HIDES BUGS` x86 already gives release/acquire for ordinary loads and stores, so wrong orderings often "work" on a developer's PC and fail on ARM (phones, Apple silicon, many servers). The table of outcomes above shows what the **C++ model allows**, not what x86 does.

- Both sides must take part: a release store read by a **relaxed** load publishes nothing.
- A **fence** (`std::atomic_thread_fence`) gives the ordering without tying it to one variable. Know it exists; reach for acquire/release on the atomic first.
- Without any synchronization, two threads touching the same non-atomic variable (one writing) is a **data race: undefined behaviour**, not merely "stale data".

> KEY POINT "What does acquire/release guarantee?" Everything written before the release is visible after the matching acquire. Say *synchronizes-with* and *happens-before*, and that `seq_cst` is the safe default.

Ordering patterns that matter

PATTERN	ORDERS	WHY
Publish with a flag or pointer	store `release`, load `acquire`	the data written before the store is visible after the load
Statistics counter	`relaxed` `fetch_add`	only the final total matters; no other data depends on it
Reference count	increment `relaxed`, decrement `acq_rel`	adding a reference needs no ordering; the thread that drops the last one must see every other thread's writes before it deletes
Spinlock	lock `acquire`, unlock `release`	a critical section must not leak out of the lock in either direction
Shutdown flag	`relaxed` if nothing else depends on it, else `release` / `acquire`	a flag alone only needs to be seen eventually
SPSC queue indices	own index `relaxed`, other's `acquire`, publish `release`	a single-producer, single-consumer ring buffer

```
// like shared_ptr's control block
struct Counted {
    std::atomic<int> refs{1};
    void addRef()  { refs.fetch_add(1, std::memory_order_relaxed); }
    void release() {
        // the last one out sees everyone's writes …
        if (refs.fetch_sub(1, std::memory_order_acq_rel) == 1)
            // … before destroying the object
            delete this;
    }
};
```

Compiled: five owners on four threads, the object was destroyed exactly once; a 2,000-round release/acquire publish test read a wrong value 0 times.

Double-checked locking (the classic trap)

```
std::atomic<Widget*> inst{nullptr};
std::mutex m;
Widget* get() {
    // 1st check, no lock
    Widget* p = inst.load(std::memory_order_acquire);
    if (!p) {
        std::lock_guard lk(m);
        // 2nd check, under the lock
        p = inst.load(std::memory_order_relaxed);
        // publish a FULLY built object
        if (!p) { p = new Widget; inst.store(p, std::memory_order_release); }
    }
    return p;
}
```

Without the release/acquire pair, another thread could see the pointer before the constructor's writes. In practice use a function-local `static` or `std::call_once` and let the compiler do this.

> **BE HONEST ABOUT WHAT YOU KNOW** Start with a mutex or `seq_cst`, measure, and weaken orders only when you can say which happens-before edge each one provides. "I use acquire/release for publication, relaxed for counters" is a sound rule of thumb.

Thread-safe data structures

Three levels, in the order you should try them:

LEVEL	HOW	EXAMPLE
1. One coarse lock	a mutex around the whole structure	a stack or map with a mutex (see the interface races in Multithreading)
2. Fine-grained locks	different mutexes for different parts, so threads that touch different parts do not wait	a queue with a head mutex and a tail mutex; a hash table with one mutex per bucket
3. Lock-free	atomics and CAS loops, no mutex	a Treiber stack, an SPSC ring buffer

Fine-grained: a queue with two locks

```
template <typename T> class TwoLockQueue {
    struct N { std::shared_ptr<T> data; std::unique_ptr<N> next; };
    // producers take tailM_, consumers take headM_
    std::mutex headM_, tailM_;
    std::unique_ptr<N> head_;
    N* tail_;
public:
    // a DUMMY node: head and tail never meet on the same node
    TwoLockQueue() : head_(new N), tail_(head_.get()) {}
    // locks tailM_ only
    void push(T v);
    // locks headM_ only (and tailM_ briefly to read the tail)
    std::optional<T> tryPop();
};
```

The **dummy node** is the trick: with it, a push only touches the tail and a pop only the head, so one producer and one consumer never block each other. Compiled: a producer pushing 1 to 1000 and a consumer popping them summed to 500500.

Read-mostly: a lookup table

A price table read by many threads and updated rarely: `std::shared_mutex` with `shared_lock` for readers and `unique_lock` for writers (Other mutex types). Return **copies** or small values, never references into the table.

Lock-free: a stack push

```
struct Node { int value; Node* next; };
std::atomic<Node*> head{nullptr};

void push(int v) {
    Node* n = new Node{v, head.load(std::memory_order_relaxed)};
    while (!head.compare_exchange_weak(n->next, n, std::memory_order_release,
std::memory_order_relaxed)) {}
        // on failure compare_exchange_weak reloads n->next with the current
    // head: just retry
}
```

Compiled: four threads pushing 1000 nodes each gave 4000 nodes, no lock. **Pop is the hard part**: another thread may pop and delete the node you are reading (**memory reclamation**), and the ABA problem can corrupt the list. Real implementations need hazard pointers, epochs or a counted pointer.

> **LOCK-FREE IS NOT FASTER BY DEFAULT** A contended CAS loop can be slower than a mutex, and a bug shows up once a month on a customer machine. Use a mutex first, measure, and use a proven library (or the SPSC ring when one producer and one consumer is enough) before writing your own.

- **Wait-free** (every thread finishes in a bounded number of steps) is stronger than **lock-free** (some thread always makes progress) which is stronger than **obstruction-free**.
- Design the **interface** first: whole operations (`tryPop` returning an `optional`), not `empty()` then `pop()`.

Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
Atomic operations and compare_exchange	Effective Modern C++	Item 40, Use std::atomic for concurrency, volatile for special memory	289 to 297
	C++ Concurrency in Action	5.2 Atomic operations and types in C++ (atomic_flag, atomic<bool>, pointers, integers, free functions)	151 to 164
	C++ Concurrency in Action	7.1 to 7.3 Lock-free data structures: definitions, a lock-free stack, memory reclamation, the ABA problem	229 to 274
Memory orders and happens-before	C++ Concurrency in Action	5.1 Memory model basics (objects, memory locations, modification orders)	148 to 150
	C++ Concurrency in Action	5.3 Synchronizing operations and enforcing ordering (synchronizes-with, happens-before, the memory orders, release sequences, fences)	165 to 196
Ordering patterns that matter	C++ Concurrency in Action	5.3 Synchronizing operations and enforcing ordering (synchronizes-with, happens-before, the memory orders, release sequences, fences)	165 to 196
Thread-safe data structures	C++ Concurrency in Action	6.2 Lock-based concurrent data structures (a thread-safe stack and queue, fine-grained locks)	199 to 216
	C++ Concurrency in Action	7.1 to 7.3 Lock-free data structures: definitions, a lock-free stack, memory reclamation, the ABA problem	229 to 274